supernova spectaculars hosted by our galaxy—one in 1572 and another in 1604—yet their wondrous visible light was widely reported.

The range of wavelengths (or frequencies) that comprise each band of light strongly influences the design of the hardware used to detect it. That's why no single combination of telescope and detector can simultaneously see every feature of such explosions. But the way around that problem is simple: gather all observations of your object, perhaps obtained by colleagues, in multiple bands of light. Then assign visible colors to invisible bands of interest, creating one meta, multi-band image. That's precisely what Geordi from the television series *Star Trek: The Next Generation* sees. With that power of vision, you miss nothing.

Only after you identify the band of your astrophysical affections can you begin to think about the size of your mirror, the materials you'll need to make it, the shape and surface it must have, and the kind of detector you'll need. X-ray wavelengths, for example, are extremely short. So if you're accumulating them, your mirror had better be super-smooth, lest imperfections in the surface distort them. But if you're gathering long radio waves, your mirror could be made of chicken wire that you've bent with your hands, because the irregularities in the wire would be much smaller than the wavelengths you're after. Of course, you also want plenty of detail—high resolution—so your mirror should be as big as you can afford to make it. In the end, your telescope must be much, much wider than the wavelength of light you aim to detect. And nowhere is this need more evident than in the construction of a radio telescope.

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Radio telescopes, the earliest non-visible-light telescopes ever built, are an amazing subspecies of observatory. The American engineer Karl G. Jansky built the first successful one between 1929 and 1930. It looked a bit like the moving sprinkler system on a farmerless farm. Made from a series of tall, rectangular metal frames secured with wooden cross-supports and flooring, it turned in place like a merry-go-round on wheels built with spare parts from a Model T Ford. Jansky had tuned the hundred-foot-long contraption to a wavelength of about fifteen meters, corresponding to a frequency of 20.5 megahertz.[††††] Jansky's agenda, on behalf of his employer, Bell Telephone Laboratories, was to study any hisses from Earth-based radio sources that might contaminate terrestrial radio communications. This greatly resembles the task that Bell Labs